

Title

Energy Conversion Effect: A Universal Law of Matter and Energy in the Cosmic System

Abstract

The traditional concept of "energy consumption" has long restricted human cognition of matter–energy interaction. Based on the Unified Theory of Cosmic Minimum Matter Units and the logical boundary of human observation, this paper proposes the Energy Conversion Effect (ECE) as a universal fundamental law. The core conclusion is: there is no absolute energy consumption in the cosmic material system; all energy activities of matter are essentially the directional transformation from one energy form to others, accompanied by adaptive evolution of material structure and ordered transmission of energy information. This paper defines the scientific connotation and core characteristics of ECE, explains its consistency with the laws of thermodynamics, and verifies its universality from microscopic particles, cosmic matter systems to living organisms. Furthermore, ECE reveals the physical essence of diseases as disordered energy conversion in life systems, and provides a unified physical foundation for reconstructing life science and modern medicine. ECE breaks the cognitive bias of traditional energy consumption theory, forms a complete closed loop with the Unified Theory and Life Conduction System Theory, and has important theoretical value and interdisciplinary guiding significance.

Keywords

Energy Conversion Effect; ECE; Universal Law; Matter–Energy Unity; Cosmic Minimum Matter Unit; Life Energy Metabolism; Energy Homeostasis; Systems Medicine